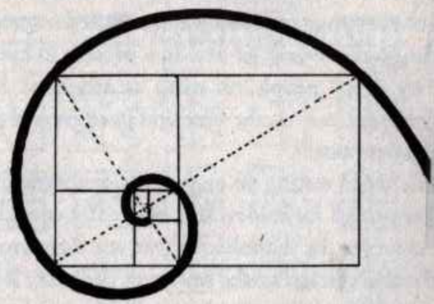
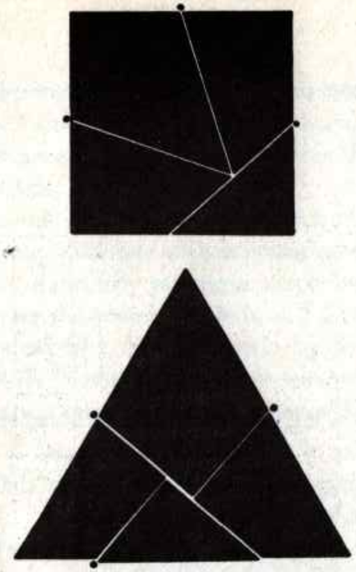
The Square

As broad and as high a man standing with outstretched arms, since the times of the earliest writings and in the earliest stone engravings the square has stood for the idea of the enclosure, the house, the village.

Enigmatic in its simplicity, in the monotony of its four equal sides and four equal angles, it generates a whole series of interesting figures: a group of harmonious rectangles, the Golden Section and the logarithmic spiral which is found in nature in the organic growth of many forms of life.



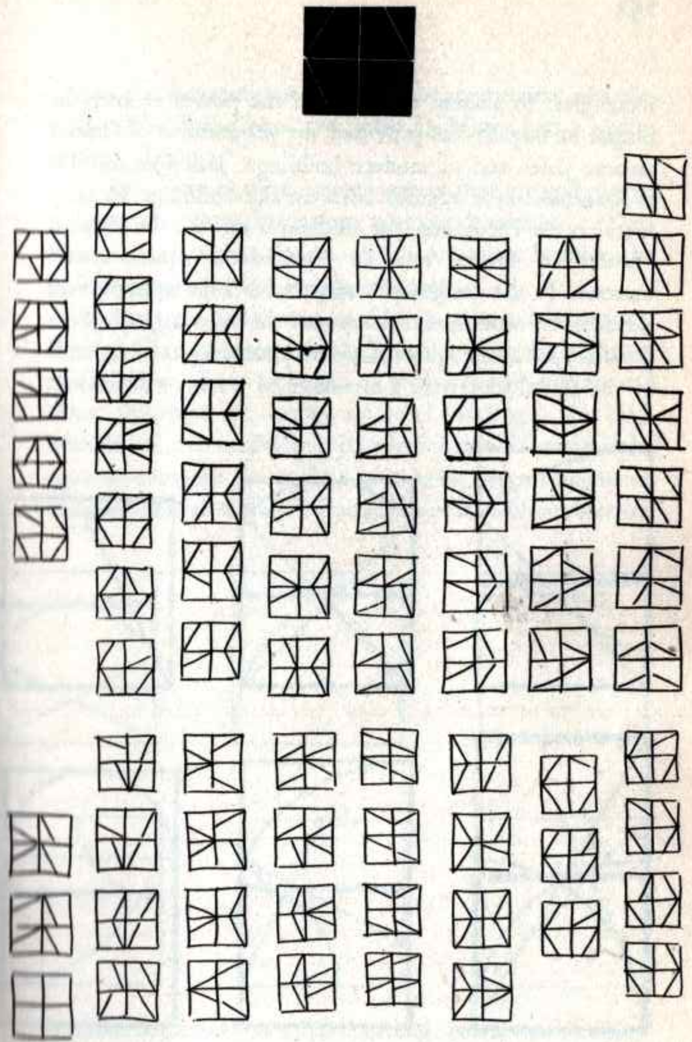
The logarithmic spiral is based on a series of 'golden' rectangles arranged in progression around the smallest of them.



A square cut as shown in the above diagram can be turned into an equilateral triangle simply by turning the pieces round as on hinges at the points marked.

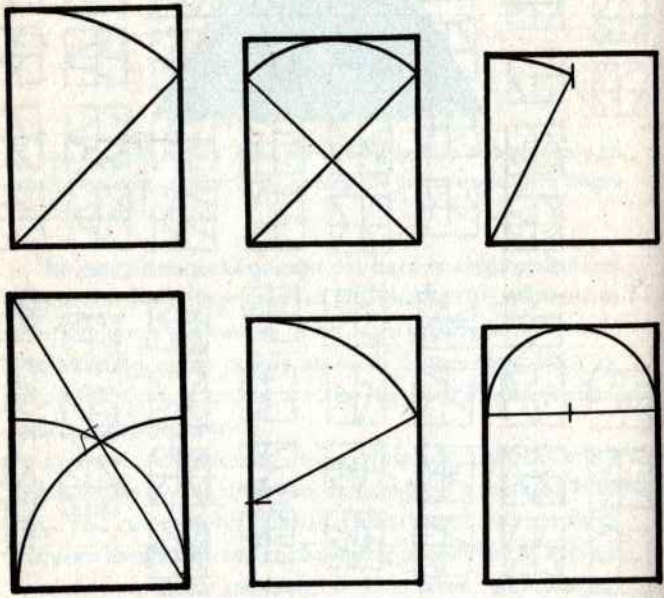
Its many structural possibilities have enabled artists and architects of every age to give a well-produced framework to their buildings and works of art. It is present in every style produced by every people on earth at any time, both as structural element and as the base and background of particular decorative motifs.

It is static when resting on one of its sides, dynamic when balanced on one of its angles. It is magic if it contains numbers, and can even be diabolic if these numbers are related between themselves and to the square or the cube. It is found in nature in many minerals. It is a curve, according to Paeanus. By cutting it up and putting the pieces together in a different arrangement you can make it into triangles or



A square cut as shown by the white lines on the black figure at the top can be put together again in many different ways.

rectangles. In ancient times it had the power to keep the plague at bay. It has provided the proportions of famous ancient cities and of modern buildings: Babylon and Tel el-Amarnah were square, while of the buildings we may mention the Parthenon, the Cathedral at Pisa, the Palazzo Farnese in Rome, and Le Corbusier's 'square spiral' museum. In the designs of many churches the square spaces beneath the semicircular domes are the most logical spaces possible, just as the square shape of a photograph corresponds to the round lens with a minimum of waste or distortion.

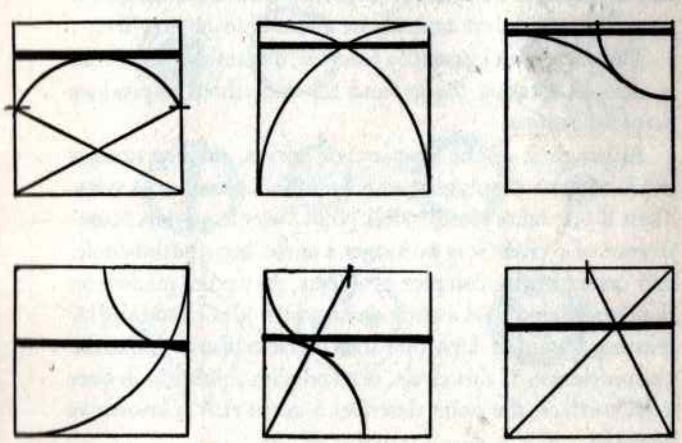


Proportioned rectangles obtained from the square by projecting its own dimensions outside itself.

On the Acropolis of Olympia the gymnasium, the *theecoleon*, the *leonidaeum* and other buildings had a square ground plan...

It has given birth to ancient games that are still played today: chess, draughts, halma, nine men's morris.... Then there are the square dances of the American frontier, the hollow squares of the British redcoats.

At the time of the eastern Chin it gave a stable square form to Chinese ideograms. It contributed to the structure of the letters of our alphabet, as well as to the Hebrew and other alphabets. A double square of matting is the basic module of the traditional Japanese house. Twenty-eight squares compose the surface of a brick. According to an ancient Chinese saying, infinity is a square without corners.



Divisions of the inside of a square starting from the main combinations obtained between lines and curves derived from the dimensions of the square itself.

The Circle

If the square is bound up with man and his works, with architecture, harmonious structures, writing and so on, the circle is related to the divine. The circle has always represented and still represents eternity, with no beginning and no end. An ancient text says that God is a circle whose centre is everywhere and whose circumference nowhere.

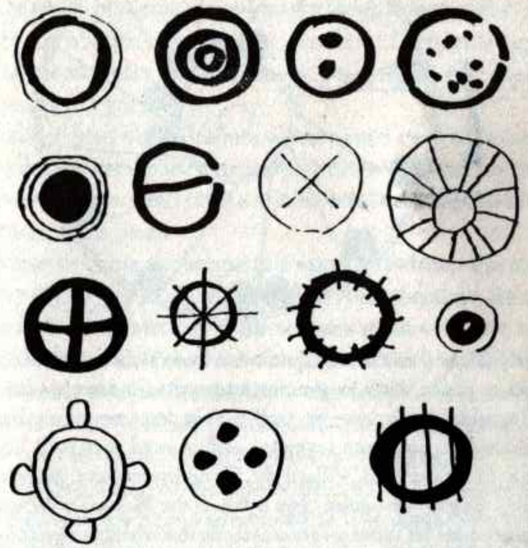
The circle is an essentially unstable, dynamic figure. From it arise all rotating things, and all vain efforts to produce perpetual motion.

Although it is the simplest of curves, mathematicians think of it as a polygon with an infinite number of sides. Then if one takes one invisible point away from the circumference of a circle it is no longer a circle, but a Patho-circle, and causes highly complex problems. Any point marked on the circumference of a circle destroys the idea of eternity by creating a point of departure and therefore also an end to the circumference. If this circle, marked with a point, rolls over a flat surface, the point describes a curve that is known as a cycloid.

It is easy enough to find circles in nature. We only have to throw a stone into calm water. The sphere, on the other hand, emerges spontaneously in soap bubbles. A section of

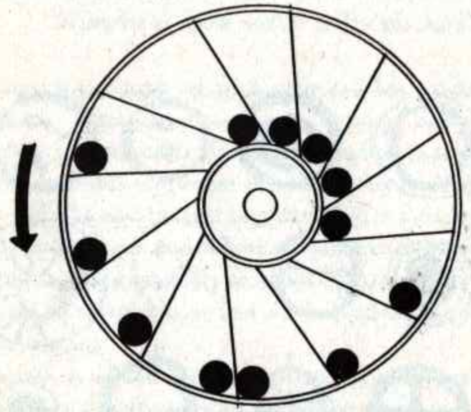
tree-trunk reveals the concentric rings which mark its growth.

A circle drawn by hand proved Giotto's mastery. A circle is always one of the first figures that children draw. When people want to gather closely round to watch something, they automatically form a circle. In this way came the form of the circus and the arena. Famous painters have used round surfaces to paint on, basing their composition strictly on the circular form. In certain cases, as in Botticelli's *Virgin with Child*, the effect of the work is spherical.



At the very beginnings of writing we find the circle in nearly all alphabets and systems of ideograms. Circles also occur often in the drawings of children below school age, in those of illiterate adults, and of prehistoric man.

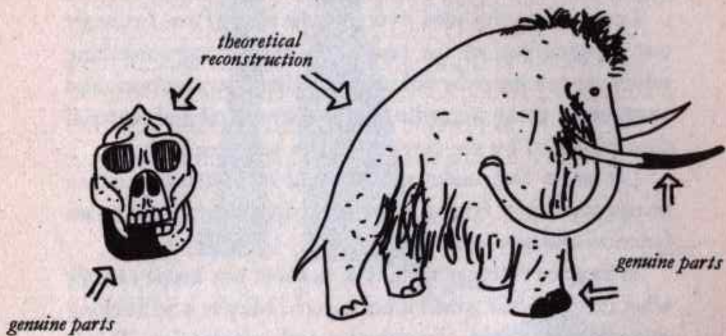
A disc put down on a flat surface can never be crooked. That is why plates are almost always round: easier to lay a table with. If they were hexagonal, square, octagonal or rectangular, they would be hard to line up neatly for dinner. And more important, we can say of a sphere that it is absolutely impossible to knock over.



Nearly all the many failed experiments in perpetual motion used the wheel or circle. Both in genuine attempts to construct a machine that would turn for ever by itself, and in demonstrations that this is impossible, unknown inventors and learned men such as Leonardo da Vinci and Villard de Honnecourt have made static models. One of the oddest cases is that of the Marquis of Worcester, who amongst his other inventions made this wheel shown above. It was supposed to rotate on its own, thanks to the weight of the balls that rolled outwards along the tilted spokes and thus acquired more leverage than the balls near the axle.

The ingenious Marquis later became famous for his piquant sauce.

Theoretical Reconstructions of Imaginary Objects



Every now and then archaeologists digging in the Sahara, or in some cave that was once on the sea shore, find a fragment of animal remains. By close examination they discover that it is a bit of the tooth of a creature that lived in the Upper Palaeolithic Age, some hitherto unknown species of Man.

This fragment passes into the hands of other experts, who try to reconstruct the whole animal, man or object (as the case may be) on the basis of structural measurements and analysis of the material and so on.

We see a lot of these reconstructions in Natural History museums, especially of course in the departments dealing with life on our planet in remote eras of which we know little or nothing. In other departments we see vases reconstructed from tiny scraps of pottery found in some tomb, and if there is a design on these an effort is made to reconstruct this as well as the pot itself.

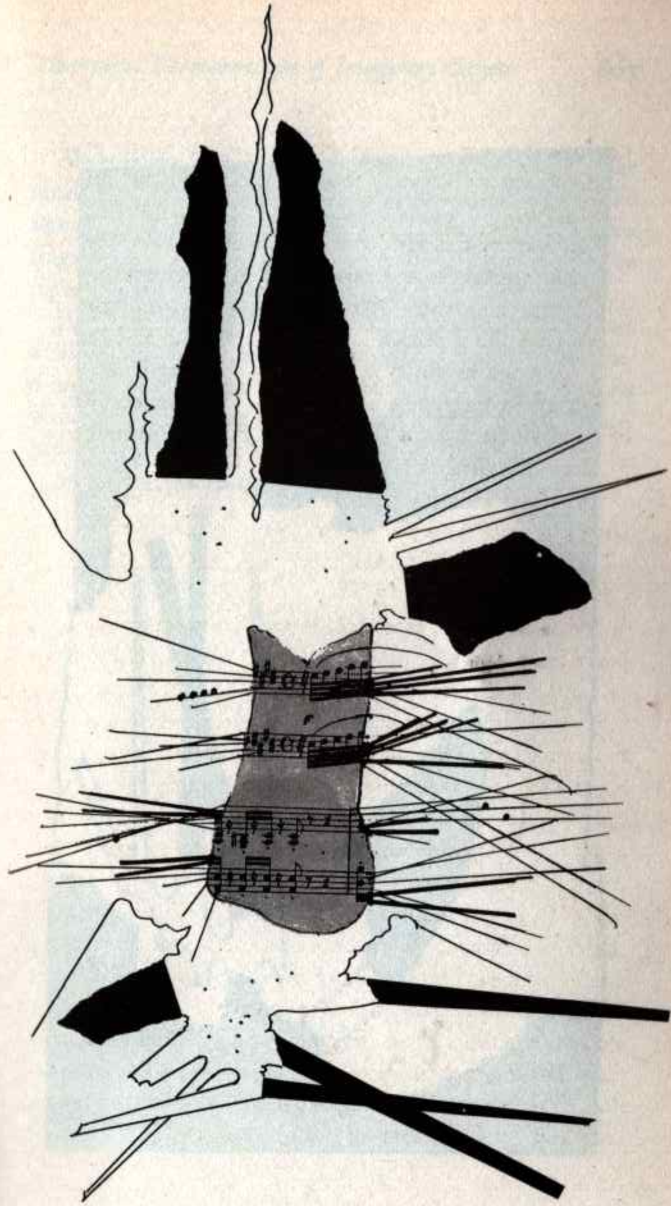
As everyone well knows, the genuine part is left just as it was found while the reconstructed parts are made of quite different materials, partly to make the reconstruction work stand out.

Let us carry this idea over into the field of art. Let us set our imaginations to the task of reconstructing something which we assume to be unknown and build up a fantastic and unexpected thing according to the structural and material data provided by the few fragments we have to go on.

Let us in fact make a theoretical reconstruction of an imaginary object, basing our work on fragments of unknown function and uncertain origin.

Whatever emerges from this, we will not know exactly what it is, or what world it belongs to. Maybe it will belong solely to the world of aesthetics and imagination. This is how we do it.

We take a few scraps of black paper, or coloured paper, or paper from a packet, wrapping paper, a sheet of music, a rag, or anything else that comes to hand. We tear the first of these into two or three pieces and drop the pieces on to a sheet of drawing paper. Then we go on to the next kind of paper. The objects (they are nothing less than the fragments we have discovered) will fall on to the paper any old how. We then look at them for quite some time, and maybe we will want to move something, but we must not do this according to any rule of logic, but simply (as Hans Arp said) according to 'the rule of the moment'. We must 'feel' something that makes our hand move. Having made any required changes of this kind we go on to join up the various fragments. To do this we must study the outlines of the fragments and their internal structures. If a fragment is torn it has a different outline from one that is cut, so that torn fragments will be joined by ragged lines and cut fragments by straight ones.



Sheet music is marked with the lines of the stave and the notes, and we assume that these will behave rather like the threads of a torn piece of cloth, only they will be rigid. If a fragment is stained or marked in any way, the stains will be reproduced also in the reconstructed part.

And thus, slowly and without thinking of Raphael, we shall have reconstructed something that never existed, something no one has ever seen, something that even we had no inkling of, something we will at once bung into the wastepaper basket because it is a mess.

If at first you don't succeed. . . .

Exercises in Topology, or Rubber-Sheet Geometry

'The atomic age in which we live is beginning to change our ideas about reality. The darkness of the unknown world is being illuminated. Visions at one time incompatible with the concept of reality are now no longer imaginary. From here there will arise new forms of visual art.

Science creates new means of knowledge. But the new inquiry is simply a return to the curiosity of children inquiring into the unknown. In this sense art and science are twins. Art serves as mediator between the imagination of the child and the adult's sense of reality. The fate of visual art in the atomic age will depend on the way in which these tendencies of mind unite.' (*Felix Deutsch*)

Geometrical figures and solids are bodiless, abstract and perfect. In geometry a cube of lead and a cube of sponge are both of them just cubes, and if their sides are of equal length the two cubes are equal, whatever they are made of.

But topology gives these forms material being, and wants to know what would happen to a tetrahedron if all its sides were sheets of rubber joined together. Topology opens the form up, shoves its hands inside, pulls it about as a furrier does the skin of an animal. Topology breaks form down on to a plane surface and finds that a tetrahedron reduced to two dimensions is a triangle. In geometry a circle is the place of residence of all the points that are equidistant from a given point. A circle with some points nearer the centre than others is no longer a circle.

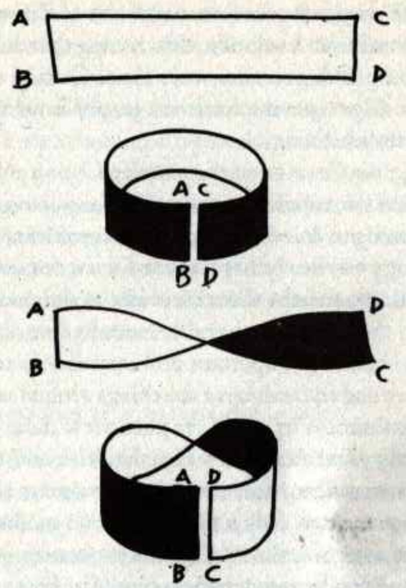
In topology a circle might be made up of the links in a chain, rather than just a series of points. One can push it about as much as one wants to, as long as it retains the essential feature of dividing space into two parts, one inside the figure and one outside. It can have any shape you like, even a square. In topology this has the same value as the circle.

The prince of geometrical instruments is the pair of compasses. The tops in topology is the pair of scissors. Topology cuts things up, puts them together again in a different way and says they are still the same, or that their qualities are equal, or that they are completely absurd like the famous Möbius Strip. Is it possible that a plastic body has only one surface? That it has no inside or out? That it is confined by a single line?

Let's see. Take a paper strip two inches wide and eight or ten inches long (the exact measurements don't matter). Place the two ends together so that the edges touch. You will have a ring, or (looked at another way) a cylinder two inches high. Open the ring again and twist one end round 180 degrees. Now place the two ends together as before, and what do you see? That what was the inside of the cylinder is now, for half the circumference, the outside, and that the two lines which previously formed the top and bottom of the cylinder are now one single and continuous line. Now does that seem right to you? But it's not all. . . .

Things were complicated still further by a certain Mr Klein, who produced a kind of bottle with a neck that turned back into the bottle through one of the sides and then came out at the bottom, joining up with the outside of the bottle. But in point of fact this bottle *has* no inside or outside. It is all one continuous surface.

These exercises and many others arose from a brain-twister posed to a bunch of good-for-nothings idling their time away in a café in Königsberg during the first half of the eighteenth century.



The Möbius Strip.

Now it happened that the River Pregel flowed through that ancient university city, and that in the middle of the river there were two islands. Joining the islands to each other and to the river banks were seven bridges. It was raining. One of the idlers said: 'One afternoon in March a tramp wanted to take a walk across all seven bridges without crossing any of them twice. How could he do it?'

There was an embarrassed silence. They ordered another round of beer, and several hours later they all went off about their business without having spoken a single word. But the problem had been squarely put, and slowly it made its way around the world. It was Leonard Euler of Petersburg who came forward with a solution, discovering that the thing was possible only if the vertices were even. Casting the islands and their bridges into the form of a graph, Euler was able to demonstrate his theory.

In 1847 the German mathematician Listing published the first treatise on topology, and today any geometry book worth looking at has some topological problems at the end.

Topology obviously has a great future, for a few happy insights, as apparently absurd and idle as the ones about the tramp and the bridges or the rubber tetrahedron, can give rise to a great number of important problems that in turn help us to a deeper understanding of the things around us and their transformations. Why don't we pose a few more problems? We are only prevented by the idea that it is not a serious matter. It is not serious, for example, to consider the thermic aspect of geometry. Why not? If we can meditate on the transformations of a rubber cube or a paper strip, why can we not also wonder about the changes caused by heat or anything else? What happens to a cube at 20 °Centigrade? At 50 °? At 100 °? What does it become?

EXPERIMENTS

1. A geometrical form exposed to intense heat loses a dimension

Take a cube of lead each side of which measures one twenty-fifth of an inch and place it in the centre of a vast horizontal iron sheet. Heat the iron gradually to *x* degrees and then let

it cool. Watch how the cube is altered by the heat. The corners gradually become round, first the lower and then the upper ones. The base expands, its corners getting increasingly rounder. They will never become square again. The cube then looks like a squat rounded pyramid, and finally a large flat disc.

Is there a relationship between the radius of the circle thus formed and the original side of the cube? Between the area of the circle and that of one side of the cube?

The experiment is irreversible. If we heat up the leaden disc again we do not produce a cube.

2. Transformation of a triangle into a sphere

Take some Plasticine and roll it out flat as housewives roll out pastry. Cut out an equilateral triangle with each side about eight inches long. Fold the three angles in towards the centre of the triangle until they touch. You have a hexagon. Make three more folds, following the diagonals, and you have a triangle again. Repeat this operation for as long as the material permits. When folding is no longer possible hold the object in the palm of one hand and, exercising a light pressure, move the other hand in a rotary manner upon it until the object has become a sphere.

This experiment clearly shows that the sphere weighs the same as the triangle. What we have to do is establish the relation between the sides of the triangle and the maximum circumference of the sphere, between the apothem of the triangle and the radius of the sphere, and between the area of the triangle and that of the sphere.

It seems most likely that $x = 3a + (6E + :2) \div r2 + aT + bT + cT - 1$, or $X = 3t - 1$.

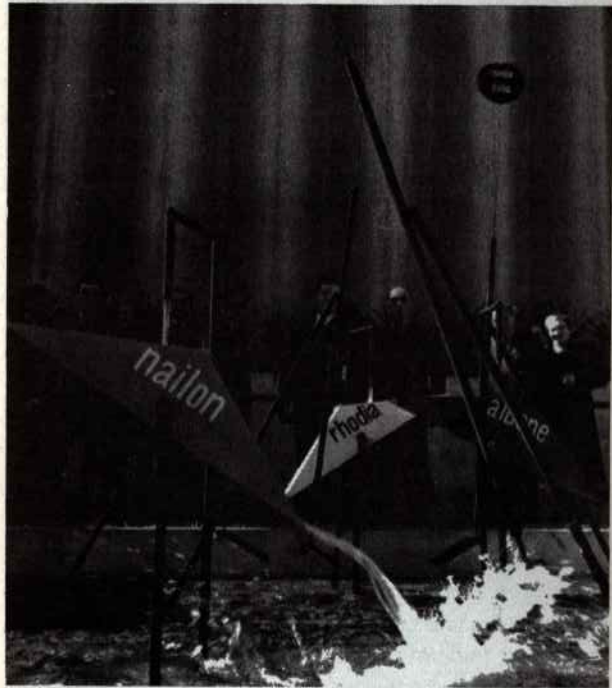
3. The fourth dimension of a cylinder

Take a cylindrical receptacle and fill it with H_2O. What you will have is a cylinder of water. Raise the temperature of this cylinder to 100° Centigrade, and you will observe the progressive reduction of one of the dimensions and one only: the height lessens, while the other dimensions remain the same. At the moment when the height of the cylinder of water has been reduced to zero you will have the complete transformation of a three-dimensional solid into a dynamic fluid mass with four dimensions. This experiment holds good for any solid form obtainable from a liquid. The form resulting from such an operation is incapable of recuperation and cannot be measured, like all forms that have a fourth dimension.

4. Transformation by abrasion

Take a series of geometrical solids made of the same material and ranging from the tetrahedron to the icosahedron. Put them in one of those rotating cylinders that are in fact specially made for rounding off objects with unwanted protuberances. It will be interesting to see how long it takes to rub off all the corners and reduce all the objects to spheres; to see the time-relation of the sphere derived from the tetrahedron to that derived from the icosahedron; to discover the relationship between the material and the form; to find out who on earth would make such calculations and how much he would ask for the job.

Two Fountains, Nine Spheres



Designs for fountains based on the principle of the water-meter: a double-triangular container, as shown in the photo, is held in balance on a supporting structure. A trickle of water falls from a hole in the tube above, filling one half of the container. When the weight of the water overcomes the weight of the empty half, the container tips over and pours the water out. Meanwhile the other half slowly fills up until. . . .

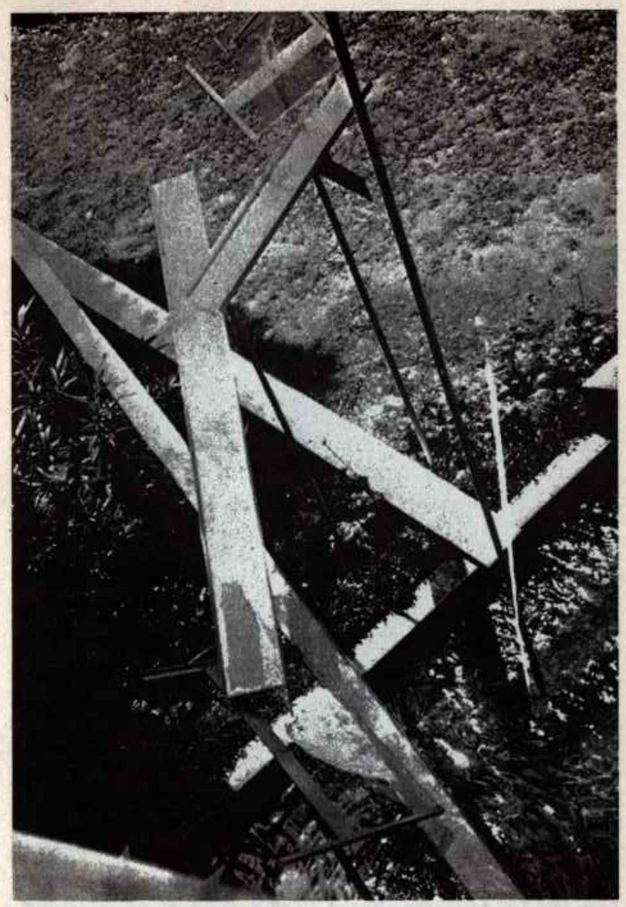
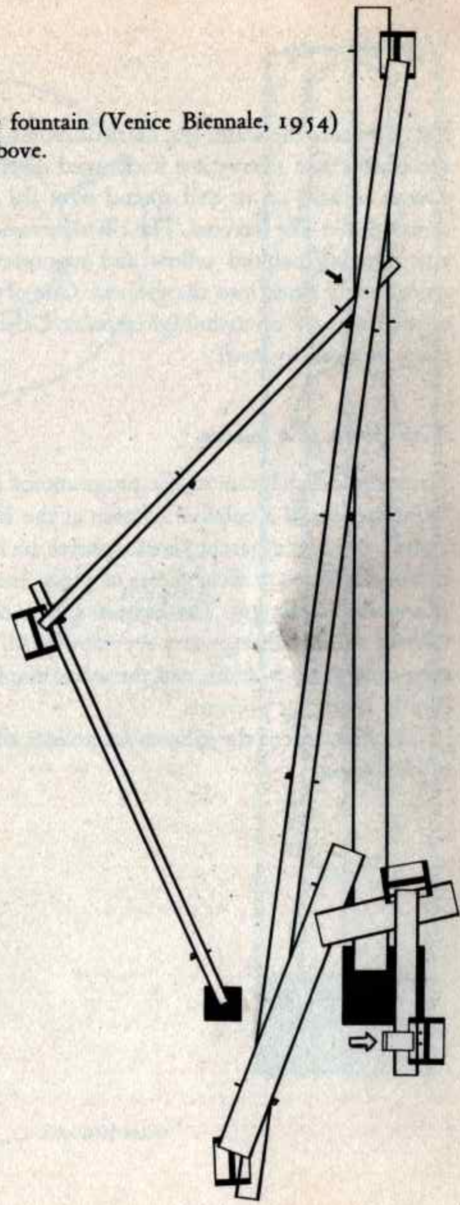


Photo and plan of the first fountain designed in 1954 and erected in front of the Book Pavilion at the Venice Biennale Exhibition.

A series of slightly tilted water-shoots started from a height of rather over six feet above the ground and eventually discharged into a black hole. During its long journey from

The waterslide fountain (Venice Biennale, 1954) viewed from above.



top to bottom the water was carried across a lawn, over some shrubbery, and every time it changed direction it splashed almost silently on to and spread over the surface of glass sheets rather like lecterns. The tilted runnels were made of zinc sheeting painted yellow and supported by tubes that were simply stuck into the ground. One of these tubes conveyed the water up to the highest point. Otherwise the whole thing worked by itself.

Nine spheres in a column

Drawing and diagram of the programmed art object called 'Nine spheres in a column' (shown at the Mostra Olivetti, 1962). Nine transparent plastic spheres are held in a vertical column by three vertical sheets of glass. Inside each sphere is a white diaphragm. The bottom sphere rests on the turntable of a motor that gyrates very slowly. All the spheres thus turn slowly, by friction, and the white diaphragms are constantly changing position.

The diameter of the spheres determines all the dimensions of the object.